



PROJECT PROPOSAL

Cyber Kill-Chain Simulator: Red vs Blue Live Defense Platform

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About Us

This project delivers a self-contained cyber range that simulates a live attack-and-defense scenario on a single laptop. An automated Red Team launches real reconnaissance and exploitation techniques against a deliberately vulnerable target, while a Blue Team detection stack logs and alerts on that activity in real time – all visualized on a live security dashboard.

It demonstrates, hands-on, the complete offense-defense loop that professional security teams work with every day: attack, detect, and respond.



Project Overview

A portable, one-command cyber range that models a real Red vs Blue engagement – automated attacks, automated detection, and a live dashboard – using industry-standard tools rather than toy simulations.

Target Audience:

Cyber security course students and instructors

Anyone learning offensive (Red Team) or defensive (Blue Team) security

Employers/professors evaluating practical, hands-on security skills



Objectives



Simulate
realistic attack

Detect and
log that
activity

Visualize
attacks

Understand
core
penetration-
testing

Understand
SIEM concept

Docker-based
service
orchestration



TECHNOLOGY STACK

Attack Tools (Red Team):

- Kali Linux — attacker OS image
- Metasploit Framework — exploitation
- Nmap — reconnaissance & service scanning

Detection Stack (Blue Team):

- Wazuh Manager — rule-based detection engine
- Wazuh Indexer — log storage & search (OpenSearch-based)
- Wazuh Dashboard — live alert visualization

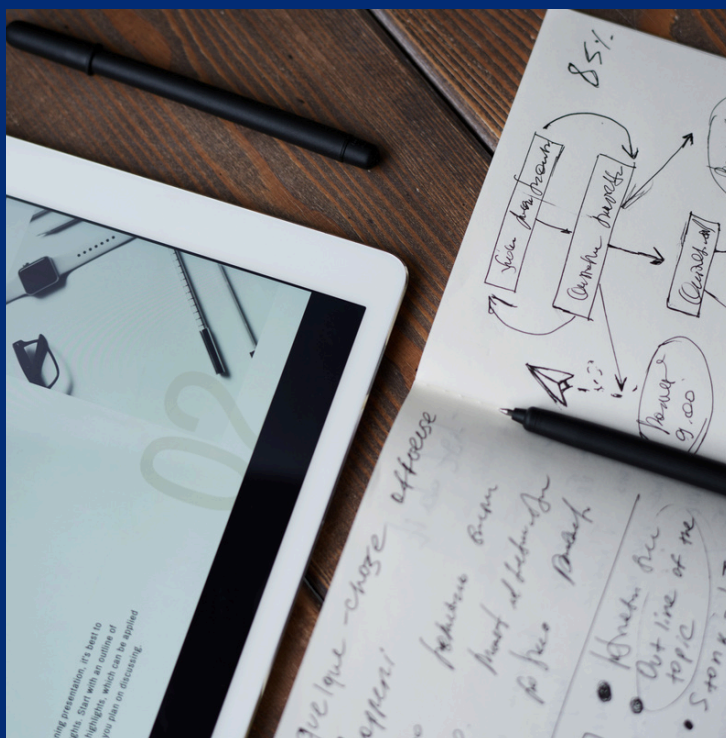
Infrastructure:

- Docker & Docker Compose — one-command orchestration
- DVWA — deliberately vulnerable target application

Case Studies

Future Scope

- Add more attack scenarios (privilege escalation, lateral movement)
- Add custom Wazuh detection rules tailored to each attack step
- Extend to a multi-host setup for a more realistic network topology
- Add automated reporting that summarizes each attack/detection cycle



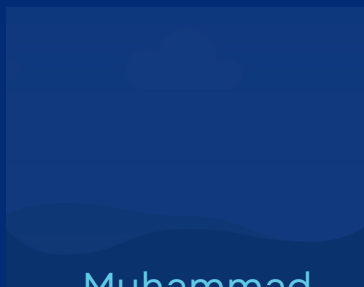
Team Expertise



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Project Timeline

A vertical timeline diagram consisting of three dark blue circles connected by a light blue vertical line. Each circle contains a phase name and a duration. To the right of each circle is a list of tasks for that phase.

PHASE 1

Weeks 1

| Week 1 | Environment setup, Docker Compose skeleton | Working base network |

PHASE 2

Weeks 2

Red Team container + attack scripts | Automated scan/exploit run |

PHASE 3

Weeks 3

Blue Team stack + detection rules | Alerts firing on attack Dashboard wiring, testing, presentation | Live demo + final report |



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